

ER. SHISIR G.C.

Mechanical Engineer • Innovator • Entrepreneur

Young Scientist — awarded by the Government of Nepal

Role on Team	Green Hydrogen Project — Mechanical & R&D Lead
Based In	Butwal, Rupandehi, Lumbini Province, Nepal
Experience	10+ years in mechanical R&D, energy systems, and innovation
Specializations	Alternative fuels, gas plant systems, hydropower design, propulsion R&D
Languages	English, Nepali, Hindi

PROFESSIONAL SUMMARY

Er. Shisir G.C. is a Nepali mechanical engineer, inventor, and entrepreneur with more than a decade of hands-on experience spanning alternative fuel research, hydropower design, automotive engineering, defense R&D, and large-scale gas plant deployment. He is the Founder & CEO of G.C. Research & Development Udhyog, a Consultant Engineer in the R&D Department of the Nepal Army Headquarters, and the CTO of Gurkha Watch. He has been formally recognized by the Government of Nepal as a Young Scientist, and has received national-level honors from the Prime Minister and the Minister of Science and Education.

His professional focus sits at the intersection of clean energy, indigenous innovation, and import substitution — building technologies that reduce Nepal's dependence on foreign fuel and equipment. This focus makes him a natural fit for the Green Hydrogen Project, where his prior work on water-based fuel systems and oxygen plant deployment translates directly into electrolyzer integration, gas handling, and end-use combustion applications.

ROLE IN THE GREEN HYDROGEN PROJECT

Within the Green Hydrogen Project, Er. Shisir G.C. contributes across three converging streams of his expertise — electrolysis-based fuel R&D, gas plant engineering, and end-use combustion application — bridging hydrogen production, storage, and deployment.

Key Contributions

- Electrolyzer & water-splitting R&D** — brings direct, hands-on experience with on-vehicle electrolysis developed during his Hydro Fuel programme, which forms the foundational chemistry for green hydrogen production.

- **Gas plant infrastructure** — drawing on the deployment of 20 oxygen plants across Nepal during COVID-19, leads planning for compression, storage, piping, and safety systems applicable to hydrogen gas handling.
- **End-use integration in mobility & power** — designs adaptation kits and combustion-side modifications so green hydrogen can be deployed into Nepal's existing vehicle and generator fleet without full powertrain replacement.
- **Indigenous manufacturing** — through G.C. Research & Development Udhyog, oversees the local fabrication of prototypes, supporting in-country production rather than imported subsystems.
- **Government & institutional liaison** — leverages active relationships with the Nepal Army R&D Department and recognition from the Prime Minister's office to fast-track approvals, funding, and partnerships.

Strategic Value to the Team

Beyond technical execution, Er. Shisir G.C. brings three strategic advantages that are difficult to replicate: a track record of taking a benchtop idea to a deployed national-scale system (oxygen plants), credibility with ministries and the armed forces, and a demonstrated ability to lead multidisciplinary teams — from automotive workshops to satellite rocketry.

CORE TECHNICAL EXPERTISE

- **Alternative & Clean Fuels** — hydrogen and water-based fuel systems, HHO/electrolysis cells, fuel-blend engine tuning.
- **Internal Combustion Engines** — diesel and petrol engine overhaul, performance tuning, emissions reduction, particulate filter cleaning.
- **Hydropower Engineering** — design of micro and small-scale hydropower components from his tenure at Mega Hydropower and Engineering.
- **Gas Plant Systems** — oxygen plant sizing, installation, and commissioning across diverse geographies; transferable to hydrogen generation and storage.
- **Aerospace & Propulsion R&D** — Chief Technical Officer for Nepal's indigenous rocket program; Spaceport America Cup–selected team.
- **R&D Project Management** — leading multi-stakeholder programs across government, military, academic, and private partners.
- **Entrepreneurship & Commercialization** — Master's-level training in entrepreneurship; founder of an active R&D enterprise.

EDUCATION

- **Master's in Entrepreneurship** — Swiss School of Management, Switzerland (2021).

- **Bachelor's in Mechanical Engineering** — VTU University, Bangalore, India (2016).
- **+2 Science** — Kathmandu Don Bosco College, Kathmandu (2012).
- **Schooling** — New Horizon English Boarding School, Butwal (2010).

PROFESSIONAL EXPERIENCE

Chief Technology Officer — Gurkha Watch | 2021 – Present

- Leads the technical roadmap for the Gurkha Watch brand, overseeing mechanical design, sourcing, and quality.
- Drives indigenous design language and assembly capability for a Nepal-origin timepiece.

Consultant Engineer, R&D Department — Nepal Army Headquarters, Kathmandu | 2020 – Present

- Advises the Nepal Army's Research & Development wing on engineering problems and prototype evaluation.
- Acts as a technical liaison between the military, civilian engineering enterprises, and academic researchers.

Founder & CEO — G.C. Research & Development Udhyog | 2018 – Present

- Founded a Nepal-based R&D enterprise focused on indigenous innovation in energy, mobility, and clean technology.
- Runs end-to-end product development — from concept through prototype to field deployment — including the Hydro Fuel device and a portfolio of energy projects.
- Manages a multidisciplinary team of engineers, technicians, and fabricators.

Design Engineer — Mega Hydropower and Engineering | 2017 – 2018

- Contributed to the mechanical design of hydropower components and balance-of-plant systems.
- Gained on-site exposure to Nepal's renewable-energy infrastructure ecosystem.

Chief Engineer — Amar Auto Diesel Service | 2016 – 2017

- Led the technical workshop for diesel engine maintenance, overhaul, and performance tuning.
- Built deep hands-on familiarity with combustion behavior, fuel injection systems, and emissions — the foundation for his later alternative-fuel work.

INNOVATION & PROJECT PORTFOLIO

1. Hydro Fuel in Automobiles — Inventor & Lead Engineer

A water-electrolysis-based fuel-supplementation device intended to be retrofitted onto internal-combustion engines (bikes, cars, buses, trucks, excavators, and generators), allowing them to operate on a blend reported by the inventor as 40% water and 60% petroleum.

Stated Outcomes

- Reduces fuel costs and improves mileage on conventional engines.
- Improves pickup, torque, and engine smoothness; cleans carbon deposits and clogged diesel particulate filters.
- Targets up to 40% long-run reduction in fuel demand at the device level.
- Releases oxygen as a by-product, contributing to lower local emissions.

National Significance

Designed explicitly to attack one of Nepal's largest macroeconomic drains — petroleum imports. The same electrolysis chemistry underpinning this device is the technical backbone of green hydrogen production, making it a natural on-ramp to the Green Hydrogen Project.

2. Rocket Project — Chief Technical Officer

Leading Nepal's indigenous rocket development effort with the long-term ambition of launching domestically built satellites on a domestically built rocket.

- Team selected for the Spaceport America Cup (SAC) 2022 — one of just 25 teams chosen from 175 countries.
- Currently preparing the team for the next edition of the competition, with active design and testing work underway.
- Builds Nepal's first significant base of aerospace propulsion talent.

3. Oxygen Plant Installation Programme — Lead Engineer

During the COVID-19 emergency, led the rollout of 20 medical oxygen plants across Nepal, from the eastern to the western regions, in response to acute oxygen shortages in the country's hospitals.

- Covered the full delivery cycle: site survey, plant sizing, installation, commissioning, and operator training.
- Operated under emergency timelines while maintaining safety and reliability standards.
- Established a direct, replicable playbook for deploying gas generation, compression, and storage systems — now being adapted for hydrogen.

4. G.C. Research & Development Udhyog — Founder & CEO

Operates as an umbrella R&D house through which Er. Shisir G.C.'s portfolio of inventions and prototypes is developed, tested, and commercialized — providing the institutional vehicle for his contribution to the Green Hydrogen Project.

AWARDS & RECOGNITION

- **Young Scientist Award** — Government of Nepal.

- **Mohandhoj Basnet Prabidhi Pragma Puraskar** — awarded by the Prime Minister of Nepal.
- **Innovative Award** — Minister of Science and Education, Nepal.
- **Best Scientific Innovation Award** — Science Minister, Province No. 5.
- **2nd Prize, National Level Bike Design Competition** — Bangalore, India (2016); awarded Rs. 1,60,000.
- **1st Prize, Best Project Award** — RR Institute of Technology, Bangalore.
- **20+ additional honors** — from various local and regional organizations across Nepal and India.

LEADERSHIP & SOCIAL ENGAGEMENT

- **Vice President, Youth for Nation** — 2020 to present; mobilizing young Nepalis around national development priorities.
- **President, Help Nepal Bangalore** — 2012 to 2016; led the Nepali student diaspora chapter in Bangalore in humanitarian and outreach work.

This social-impact track record matters for the Green Hydrogen Project: it demonstrates an established ability to convene communities, students, and institutions around a shared cause — exactly what the energy transition will require at a national level.

VISION: WHY GREEN HYDROGEN, WHY NOW

Nepal spends a substantial portion of its national budget importing petroleum fuel. At the same time, the country generates significant surplus hydroelectric capacity that is exported below market value or simply curtailed in the wet season. Green hydrogen — produced by using surplus renewable electricity to split water — is the most credible technology pathway to convert that latent advantage into a domestic, exportable, zero-emission fuel.

Er. Shisir G.C.'s career has been a steady progression toward exactly this convergence: clean hydropower generation, water-splitting fuel chemistry, gas plant deployment, and end-use combustion engineering. The Green Hydrogen Project is the natural next step where those threads come together — and where his work can deliver national impact.

CONTACT

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